

What body parts are used?

All features here are built from the **orientation vector**:

bodypart2 → bodypart16

(“front/head” → “tail”). Every column below ultimately comes from that line.

Compass & Angle glossary (used everywhere in this section)

- **Clockwise_angle_degrees**: the facing angle of bodypart2 → bodypart16, measured **clockwise** in **degrees** (0–360).
 - We bin that angle into 8 compass bins:
 - N=0, NE=1, E=2, SE=3, S=4, SW=5, W=6, NW=7.
(E.g., 95° → E → digit 2.)
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1) Rotation / Orientation (per-frame direction and its summaries)

Columns you’ll see in the final CSV

- **Clockwise_angle_degrees**
The fish’s facing direction in **degrees** (0–360).
Example: 90 = facing **East**.
- **Mean_angle_time_window_{W}**
Rolling average of Clockwise_angle_degrees over the last **W frames**.
Units: degrees.
Example: If W=150 (~5 s @30 fps) and the fish faced 80–100° most of that time, this will sit near **90°**.

Helper columns (computed but removed before saving):

- Angle_radians, Angle_sin, Angle_cos, Angle_sin_cumsum, Angle_cos_cumsum
- Compass_direction (N/NE/.../NW as text), Compass_digit (0–7)

Why removed? They’re intermediate variables used to compute other features.

2) One-hot compass encoding (current facing bin)

Columns:

- **Direction_N, Direction_NE, Direction_E, Direction_SE, Direction_S, Direction_SW, Direction_W, Direction_NW**

1 if the fish is currently facing that bin, else 0.

Example: Facing $\sim 95^\circ \rightarrow$ Direction_E = 1, the rest 0.

3) 45° direction switches & “how long we kept a direction”

These describe **changes between adjacent compass bins** and **persistence** in one bin.

Columns you’ll see:

- **Degree_shift_{W}_mean, Degree_shift_{W}_median, Degree_shift_{W}_sum, Degree_shift_{W}_std**

Rolling stats of the **absolute per-frame angle change (wrapped)** over **W frames**.

Units: radians (then aggregated).

Read it as: larger numbers $\rightarrow$ wigglier/turn-heavier movement.

- **Consecutive_ms_in_same_compass_direction**

How long (in **seconds**) the fish has continuously kept the **same** compass bin up to *this* frame.

Example: value 3.2 means it’s stayed, say, “E” for 3.2 seconds.

- **Consecutive_ms_in_same_compass_direction_zscore**

Z-score of the above (unitless), useful to spot unusually long/short holds.

- **Unique_compass_directions_in_{W}_window**

How many **distinct** bins were visited in the last **W frames** (1–8).

Read it as: exploration of headings; 1 = stayed in one bin, 5 = moved through $\sim$ five bins, etc.

Helper columns (removed before saving):

- Direction_switch (1 when the bin changes),
- Compass_digit_shifted (next frame’s bin code),
- Switch_direction_value (signed step around the 0–7 ring; e.g., +1 for NE $\rightarrow$ E).

You won’t see these in the CSV; they’re used to build the rolling metrics above and below.

4) Directional switches in rolling windows (how many & which way)

Columns:

- **Number_of_direction_switches_{W}**
Count of **bin changes** in the last **W frames**.
Read it as: “how often we changed facing bin lately.”
 - **Directionality_of_switches_switches_{W}** (*name kept to match code*)
Rolling **sum of signed steps** (+1 clockwise, -1 counter-clockwise, wrapping handled).
Read it as: positive → more clockwise turns; negative → more counter-clockwise; near 0 → balanced.
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5) N-degree direction switches (big jumps vs W frames ago)

Checks whether our **current** facing direction is roughly **90°** or **180°** from **W frames ago**.

Columns (for each W):

- **90_degree_switch_{W}**: 1 if current bin is a quarter-turn from the bin **W frames ago** (e.g., N vs E/W).
- **180_degree_switch_{W}**: 1 if current bin is opposite to **W frames ago** (e.g., N vs S).

Units: binary (0/1).

Use case: detect sharp re-orientations with a lag (not just frame-to-frame jitter).

6) Angular dispersion (steadiness of heading)

Columns:

- **Angular_dispersion**
A **vector-strength** style score (0–1) from cumulative sin/cos of the angle:
1 = very consistent heading; **0** = highly variable. (*Dimensionless.*)
 - **Angular_dispersion_window_{W}**
Rolling **mean** of Angular_dispersion over **W frames**.
Use it to compare short- vs long-term steadiness.
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Two quick examples

- **Example A (steady swimmer):**

Stays near 90° (East) for 5 s.

- Clockwise_angle_degrees ~ 90 throughout
- Direction_E = 1 mostly
- Number_of_direction_switches_150 low
- Unique_compass_directions_in_150_window ~ 1–2
- Angular_dispersion_window_150 close to **1.0**
- Consecutive_ms_in_same_compass_direction grows steadily (seconds)

- **Example B (zig-zag):**

Alternates NE↔SE every few frames.

- Many bin flips → Number_of_direction_switches_{W} high
- Degree_shift_{W}_mean higher (more turning)
- Unique_compass_directions_in_{W}_window ~ 2–3
- Angular_dispersion_window_{W} lower (closer to 0)